Daniel T. Chen

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ArXiv: http://arxiv.org/a/chen_d_4

Research Interest

Applied stochastic analysis, interacting particle systems, random graphs, large deviations, multi-agent control, optimization, and scientific machine learning.

Education

Brown University — Providence, RI
Ph.D. in Applied Mathematics (Advisor: Kavita Ramanan) — Aug 2023 – May 2028

Case Western Reserve University — Cleveland, OH
B.A. in Mathematics, B.A. in Computer Science — Aug 2019 – May 2022

Employment

Northrop Grumman — Walpole, MA
Math Modeler/Software Engineer at 2Is Solutions — Sep 2022 – Aug 2023

Awards

Outstanding student poster award — June 2026
Stochastic Networks Conference

The Lajos F. Takacs Mathematics Prize — May 2022
Case Western Reserve University

The Computer and Data Sciences Research Award — May 2022
Case Western Reserve University

Publications

In Preparation/Under Review

Daniel T. Chen, Andrew W. Eckford, and Peter J. Thomas. "Capacity of a class of fully observed multistate Poisson-type signal transduction channels." In submission (2025).

Journal Publications

Juan D. Toscano, **Daniel T. Chen**, Vivek Oommen, George Em Karniadakis. "A variational framework for residual-based adaptivity in neural PDE solvers and operator learning." *NPJ Artificial Intelligence*, 2, no. 32, (2026).

Daniel T. Chen, Zain H. Saleem, and Michael A. Perlin. "Quantum circuit cutting for classical shadows." *ACM Transactions on Quantum Computing*, 5, no. 2 (2024) 1-21.

Daniel T. Chen, Brian Doolittle, Jeffrey Larson, Zain H. Saleem, and Eric Chitambar. "Inferring quantum network topology using local measurements." *PRX Quantum*, 4, (2023) 040347.

Daniel T. Chen, Alexander G. Strang, Andrew W. Eckford, and Peter J. Thomas. "Explicitly solvable continuous-time inference for partially-observed Markov processes." *IEEE Transactions on Signal Processing*, 70, (2023) 6232-6242.

Conference Proceedings

Daniel T. Chen, Ethan H. Hansen, Xinpeng Li, Aaron Orenstein, Vinooth Kulkarni, Vipin Chaudhary, Qiang Guan, Ji Liu, Yang Zhang, and Shuai Xu. "Online detection of golden circuit cutting points." *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*, 1, (2023) 26-31.

Daniel T. Chen, Ethan H Hansen, Xinpeng Li, Vinooth Kulkarni, Vipin Chaudhary, Bin Ren, Qiang Guan, Sanmukh Kuppannagari, Ji Liu, Shuai Xu. "Efficient quantum circuit cutting by neglecting basis elements." *2023 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, (2023) 517-523.

Daniel T. Chen, Betis Baheri, Vipin Chaudhary, Qiang Guan, Ning Xie, and Shuai Xu. "Approximate quantum circuit reconstruction." *2022 IEEE International Conference on Quantum Computing and Engineering (QCE)*, (2022) 509-515.

Professional Activities

Co-president of Brown Quantum Initiative — May 2025 – present

- The BQI is a student-run research-centric initiative for quantum information science at Brown University. We host seminars, tutorials, and Hackathons to help unite students and faculty interested in quantum information sciences across departments.

Journal Referee

- Annals of Applied Probability (2025, 2026), Stochastic Processes and their Applications (2025), SIAM Journal of Control and Optimization (2025), ACM Transactions on Quantum Computing (2025), Stochastic Models (2026), Journal of Computational Physics (2025)

Talks

On the Shannon capacity of biological signal transduction — Cleveland, OH
SIAM Conference on the Life Sciences — Jun, 2026

Interacting particle on sparse, random graphs: Convergence and characterization — Providence, RI
AMS New England Graduate Student Conference — Apr 2026

A variational framework for residual-based adaptivity in neural PDE solvers and operator learning — Providence, RI
Dec, 2025
CRUNCH Seminar (Recording)

Hydrodynamic limits for interacting particles with two spatial scales — University Park, PA
2025 SIAM New York-New Jersey-Pennsylvania Section Conference — Nov, 2025

The capacity of a class of fully-observed, multistate Poisson-type signal transduction channel — Ann Arbor, MI
Jun 2025
International Symposium on Information Theory 2025

Distributed quantum computing via circuit cutting — New Haven, CT
Yale Student Theory Day — May 2025

Continuous-time inference for partially-observed Markov processes Trieste, Italy
SIAM Conference on Uncertainty Quantification Mar 2024

Approximate quantum circuit reconstruction Bloomfield, CO
IEEE International Conference on Quantum Computing and Engineering 2022 Sep 2022

Variational Monte Carlo with conditional-value-at-risk objective function Chicago, IL
American Physical Society (APS) March Meeting 2022 Mar 2022

Virtual
Jan 2021

Teaching

Directed Reading Program
Division of Applied Mathematics (Brown University)

- Spring 2026: Universal approximation (mentee: Patrick Bai)
- Spring 2026: Lower bounds in statistical learning (mentee: Max Liu)
- Fall 2025: Randomness, competitions, and hierarchy (mentees: Jeffrey, Na)
- Spring 2025: Nonparametric Bayes (mentees: Avery, Chai, Sadia)
- Fall 2024: Multi-armed Bandits (mentees: Jennifer, Mia, Sabrina)

Teaching Assistance
Division of Applied Mathematics (Brown University)

- Spring 2025: APMA 1200: Operations Research: Probabilistic Models
- Fall 2024: APMA 1930Z: Introduction to Mathematical Machine Learning

Peer Tutor
Department of Mathematics (Case Western Reserve University)

- Introductory Calculus (MATH 121, 122, 125, 126), Elementary Differential equations (MATH 224), Linear Algebra (MATH 201, 307), Theoretical Statistics (STAT 345).

Service

Presenter at Math Slam Mar 2025, Mar 2026
Volunteer judge for the Local Math Modeling Contest Nov 2024
The Math Corps Cleveland Jan 2020 – Mar 2020